

Q1. Choose the correct answer : (24 Marks)

1) For transmission of normal speech signal the PCM channel needs a bandwidth of :

a. 64KHZ

b. 8KHz

c. 4KHz

2) The capacity of an analog communication channel with 4KHz bandwidth , $SNR_{db} = 15$ db is approximately :

a. 20000 bps

b. 16000 bps

c. 10000 bps

d. 8000 bps

3) The core of an optical fiber has :

a. A lower index of refraction than the air

b. A lower index of refraction than the cladding

c. A higher index of refraction than the cladding

d. None Of The Above

4) When the index of refraction is greater in material 1 than it's in material 2 , the velocity of propagation in material 1 compared to material 2 is :

a. Equal to OR Greater

b. Greater

c. Lesses

d. Equal

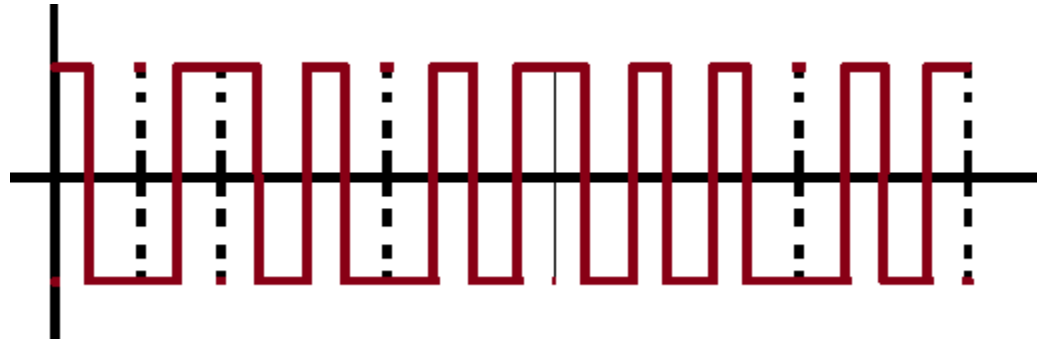
5) In the following format, how many transitions occur :

- a. 16**
- b. 14**
- c. 15**
- d. 8**

b. 14

c. 15

d. 8



6) What is one principal difference between synchronous & Asynchronous transmission :

- a. The bandwidth required a different
- b. The pulse heights are different
- c. The clocking is mixed with data in Asynchronous
- d. The clocking is derived from the data in Synchronous**

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7) A example of a bounded medium is :

- a. Coaxial Cable**
b. Wave Guide
c. Fiber Optic Cable
d. All Of The Above

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c. Fiber Optic Cable

d. All Of The Above

8) *The correctness & accuracy of transmitted message content is :*

- a. *Verified by the modem*
- b. **Determine by sender & receiver not in comm. System**
- c. *Ensured by use of digital techniques*

9) *The difference between timing AND frame is :*

- a. *Timing is concerned with the individual bits, frame is concerned with boundaries between character*
- b. *Timing refers to serial transmission , frame refers to parallel*
- c. *Timing is concerned primarily with asynchronous system , frame is concerned with synchronous system*

10) *Fiber optic cable operate at frequencies near :*

- a. *20MHz*
- b. *200MHz*
- c. *2GHz*
- d. **800THz**

11) *Transmission of binary signals requires :*

- a. *Less bandwidth than analog*
- b. **More bandwidth than analog**
- c. *The same bandwidth as analog*
- d. *A license from FCB*

12) *Codes are always :*

a. Eight bits per character

b. Either seven or eight bits per character

c. Agreed upon in advance between Tx & Rx

d. The same in all modern computers

Q2. (10 Marks) :

If we have a screen the digitize 480x500 pixels , each pixel have a 32 different value , if we sent 30 photos per second , Find Bit Rate . (5 Marks)

$$\text{Bit Rate} = 480 \times 500 \times 32 \times 30 = 230.4 \text{ Mbps}$$

If this channel have bandwidth 4.5Mhz and Signal to Noise Ratio in db is 35_{db} Find capacity Of this channel . (5 Marks)

$$\text{SNR}_{db} = 10 \text{ Log } (\text{SNR}) \rightarrow \text{SNR} = 3162.3$$

$$C = 4.5M * \text{Log}_2(1+3162.3) \rightarrow 52.3\text{Mbps}$$

Q3. (16 Marks) :

Represent (01001110) in this scheme :

- a) NRZ-L**
- b) NRZ-I (assume the previous bit is 1)**
- c) AMI (assume the previous bit 1 is represent as negative)**
- d) Manchester**
- e) Differential Manchester (assume the previous bit is 1)**